Fiscal Year:	FY 2025	**Task Last Updated:**	FY 08/20/2025
PI Name:	Lee, Stuart M.C. Ph.D.		
Project Title:	Defining the Relation Between Biomarkers of Oxidative and Inflammatory Stress and Atherosclerosis Risk in Astronauts During and After Long-Duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-2749	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Crew Health NNJ10ZSA003N
Start Date:	03/01/2016	**End Date:**	09/30/2027
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: HRP Element has been changed to the collaborating org (Space Radiation / SR), which is sponsoring the investigation per C. Ribeiro and J. Weeks / NASA HRP (Ed., 7/11/24). NOTE: End date changed to 09/30/2027 per C. Ribeiro/HHC, HRP (Ed., 3/3/25) NOTE: End date changed to 08/09/2024 per C. Ribeiro/HHC, HRP (Ed., 1/4/24) NOTE: End date changed to 12/31/2023 per C. Ribeiro/HHC, HRP (Ed., 8/26/22) NOTE: End date changed to 8/31/2023 per HRP, HHC Element (Ed., 8/12/21)		
Key Personnel Changes/Previous PI:	August 2019 report: Steven Laurie, Ph.D. is CoInvestigator. January 2017: Remove Dr. Rob Ploutz-Snyder as CoInvestigator; Add: Dr. Alan Feiveson as Collaborator.		
COI Name (Institution):	Smith, Scott Ph.D. (NASA Johnson Space Center) Feiveson, Alan Ph.D. (Collaborator: NASA Johnson Space Center) Stenger, Michael (NASA Johnson Space Center) Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center)		

Grant/Contract No.:	Internal Project
Performance Goal No.:	
Performance Goal Text:	
Task Description:	NOTE: Continuation of "Defining the Relation Between Biomarkers of Oxidative and Inflammatory Stress and Atherosclerosis Risk in Astronauts During and After Long-Duration Spaceflight" ; previous Principal Investigator was Dr. Steven Platts, until March 2016. Atherosclerosis is the major contributor to cardiovascular disease-related morbidity and mortality. Research indicates that many of the risk factors commonly associated with atherosclerosis contribute to endothelial dysfunction, a process which presents early in life before angiographic evidence of disease and precedes the clinical manifestation of many cardiovascular disease-related disorders. In an effort to compensate for the initial risk factor-related disruptions to homeostasis, there is a compensatory upregulation of atheroprotective mechanisms. However, in the absence of appropriate risk factor management, these defense mechanisms may become overwhelmed and less able to reestablish normal function. Key systems that help maintain vascular homeostasis and are susceptible to differential deleterious alterations include those that help balance levels of oxidative and inflammatory stress. New evidence suggests that long-duration spaceflight may promote oxidative and inflammatory stress through mechanisms such as radiation exposure, diet, physical inactivity, and psychological stress. However, there are no data supporting a causal link between biomarkers of oxidative and inflammatory stress and indices of vascular endothelial dysfunction in spaceflight. As such, we proposed to examine the relation between biomarkers of oxidative and inflammatory stress and well-established measures of vascular endothelial dysfunction (flow mediated dilation (FMD)) and carotid intima-media thickness (cIMT) in astronauts before, during, and after long duration spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It is well known that inflammation is a key contributor to the development of atherosclerosis. There also is emerging work showing the link with oxidative damage. This work may contribute to general clinical science by showing the interactions of multiple stressors in a unique environment (spaceflight).
Task Progress:	All pre-, in-, and immediate postflight data were collected and analyzed. A manuscript describing these results was published (Lee SMC et al. Arterial structure and function during and after long-duration spaceflight. J Appl Physiol 129: 108-123, 2020), and a response (Lee SMC et al. Reply to Greaves et al. J Appl Physiol 129: 113, 2020) to a “Letter to the Editor” (Greaves DK, Robertson AD, Patterson CA, Au JS, Hughson RL. Evidence for increased cardiovascular risk to crew during long duration space missions. J Appl Physiol 129: 1111–1112, 2020) regarding our findings also was published. Also, two papers resulting from this study acquired in the one-year mission (MacNamara JP et al. Cardiac effects of repeated weightlessness during extreme duration swimming compared with spaceflight. Circulation 143: 1533–1535, 2021) and twins study (Garrett-Bakelman FE et al. The NASA Twins Study: A multidimensional analysis of a year-long human spaceflight. Science 364: eaau8650, 2019) were published. All the astronauts who volunteered to participate in this study completed their missions more than 5 years ago, but not all were available consistently at the planned increments in the years after landing due to retirement from the astronaut corps, limited travel opportunities coming from international partner organizations, COVID-19 pandemic restrictions, and/or life events. Of the 13 astronauts who began the study, 11 participated in data collection through R+5 years, with the last data collection occurring in September 2023. Statistical analyses are in progress, and a draft manuscript describing the results of the longitudinal phase of the study is in preparation for publication in a peer-reviewed journal. The overall findings of this study are that although biomarkers of oxidative stress and inflammation are elevated during missions in low-Earth orbit, common carotid artery and brachial artery structure and function were not changed by spaceflight. Further, these do not appear to change significantly 5 years after landing. However, long-term surveillance of biomarkers, risk factors, and manifestations of cardiovascular disease is needed to quantify the long-term health consequences of spaceflight, particularly those missions with an extended duration in altered gravity fields and higher radiation exposures. Results from this study and previous reports have been used to inform the design of the NASA Standard Measures Study, as well as Human System Risk Board-defined risk, “Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes” and are included in the corresponding Human Research Program Evidence Report that was submitted to the NASA Human Research Program in 2022.
Bibliography Type:	Description: (Last Updated: 11/18/2025)
Abstracts for Journals and Proceedings	Lee SMC, Martin D, Ribeiro LC, Freeman-Perez SA, Smith SM, Zwart SR, Ferguson C, Laurie SS, Macias BR. "Indices of cardiovascular disease risk in astronauts after long-duration spaceflight in low Earth orbit." 2024 NASA Human Research Program Investigators’ Workshop Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators’ Workshop Galveston, Texas, February 13-16, 2024. , Feb-2024
Articles in Peer-reviewed Journals	Lee SMC, Ribeiro LC, Martin DS, Laurie SS, Zwart SR, Chen X, Young M, Crucian B, Smith SM, Macias BR. "Arterial structure and function in the years after long-duration spaceflight." J Appl Physiol (1985). 2025 Jun 1;138(6):1474-88. https://doi.org/10.1152/japplphysiol.00264.2024 ; PMID: 40304245 , Jun-2025